

Vishal Kumar Mathuri

BLOCKCHAIN ENGINEER & FULL-STACK DEVELOPER | SOLIDITY / EVM / RUST / SOLANA / REACT / NODE.JS

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SUMMARY

Blockchain Engineer and Full-Stack Developer with hands-on professional experience across **Solidity/EVM smart contracts, Rust/Solana programs, DeFi systems, and full-stack Web3 and web applications** — including 50+ smart contracts and 10+ full-stack blockchain applications shipped to date, plus completed full-stack projects in diagnostics and field-data workflows. Strong focus on **secure, testable engineering** using Solidity, Foundry, Hardhat, OpenZeppelin, Anchor, React.js, Node.js, Express.js, and MySQL.

Experienced across **DeFi lending and liquidation, NFTs, decentralized identity, role-based access control, Solana/Anchor program design, cross-chain interoperability (LayerZero), and production full-stack applications** — with a portfolio centered on production-oriented, security-conscious system design rather than a collection of tutorial projects.

TECHNICAL SKILLS

Smart Contracts: Solidity, ERC-20, ERC-721, access control, staking, governance, vesting, custom errors, events

EVM / Blockchain: Ethereum, EVM, Polygon, Ethereum Sepolia, gas optimization, contract verification, IPFS

Rust / Solana: Rust, Solana, Anchor, PDAs, SPL Token, Metaplex, Solana Web3.js, wallet integration

Cross-Chain: LayerZero V2, cross-chain messaging, relay architecture, message verification, replay protection

Security / Testing: OpenZeppelin, ReentrancyGuard, AccessControl, Ownable, Pausable, unit/integration/fuzz/invariant testing, end-to-end API testing

Frontend: HTML, CSS, JavaScript, TypeScript, React.js, responsive design, Progressive Web Apps (PWA)

Backend & Databases: Node.js, Express.js, REST APIs, server-side validation, MySQL, SQL, schema design, Ethers.js, Web3.js, MetaMask, Phantom Wallet

Tools: Foundry, Hardhat, Remix, Git, GitHub, VS Code, Docker, Vercel, Render, IPFS/Pinata

PROFESSIONAL EXPERIENCE

Blockchain Developer / Full-Stack Developer Jul 2025 - Present
Freelance

- Developed and deployed **50+ Solidity smart contracts** implementing ERC-20, ERC-721, staking, governance, and marketplace functionality across NFT, fundraising, identity, and access-control applications.
- Built **10+ full-stack Web3 and web applications** integrating React.js frontends, Node.js backend services, REST APIs, wallets, and EVM networks.
- Applied **OpenZeppelin security primitives** and secure design patterns including reentrancy protection and role-based access control, reducing security risk across deployed contracts.
- Debugged and tested application flows using unit and integration tests; handled frontend development, backend integration, API design, and deployment end-to-end.

AI-Web3-Full Stack Developer (Part-time) Mar 2026 - Jul 2026
EtherAuthority Remote

- Built a Telegram Mini App featuring multiple in-app mini-games, including Coinflip, Find the King (a dice-based game), and Wheel, using the Telegram Web App platform.
- Designed and implemented game logic, betting flow, and result computation for each game mode.
- Collaborated with the team on frontend/backend integration and testing to ship engaging, low-latency gameplay experiences.

Senior Content Developer Jul 2019 - Mar 2023
Think & Learn Pvt Ltd Bengaluru, IN

- Managed scheduling operations for 200+ student batches across multiple grade levels with **100% operational accuracy** and enhanced workflow efficiency.
- Improved an automated concept-tagging workflow to **95% classification accuracy** through systematic validation and quality checks, boosting system efficiency and reducing errors.
- Crafted and executed structured Learning Journeys for academic programs utilizing digital learning platforms, resulting in a **96% success rate**.
- Designed and curated interactive educational content, mock tests, and learning modules for Byju's TV and BHL P programs, increasing user satisfaction and engagement.
- Resolved student support and academic operations requests using Salesforce with **98% resolution accuracy** within 24 hours, ensuring uninterrupted course delivery.
- Collaborated with academic operations, HR, and support teams while maintaining strong cross-team communication and coordination.

FLAGSHIP PROJECTS

1. Production-Grade EVM Lending Protocol

Tech Stack: Solidity • Foundry • Chainlink • OpenZeppelin **Links:** [GitHub](#)

- Engineered a modular, overcollateralized EVM lending protocol supporting multi-asset supply, withdrawal, borrowing, repayment, interest accrual, and liquidation, with responsibilities separated across **LendingPool, ReserveManager, RiskManager**, and configurable interest-rate modules.
- Implemented index-based scaled accounting and risk management, including utilization-driven borrow/supply rates, collateral-factor borrowing limits, health-factor solvency checks, liquidation thresholds/bonuses, and rounding-aware debt/collateral accounting.
- Built a Chainlink-based oracle layer with defensive price validation, including stale-price protection, incomplete/stale round detection, and an oracle abstraction enabling deterministic mock testing.
- Validated the protocol with an extensive Foundry test suite spanning unit, integration, fuzz, and stateful invariant testing; deployed and source-verified the

contracts on Ethereum Sepolia and executed an end-to-end supply → borrow → repay → withdraw flow.

2. NFT Marketplace

Tech Stack: Solidity • ERC-721 • OpenZeppelin • React.js • Ethers.js • IPFS/Pinata Links: GitHub • Live (Vercel) • API (Render)

- Engineered a full-stack NFT marketplace supporting ERC-721 minting, listing, purchasing, and ownership transfers through Solidity smart contracts, with a React.js interface for end-to-end marketplace interactions.
- Integrated **Ethers.js** and **MetaMask** for wallet-based Web3 workflows, enabling users to connect accounts, sign transactions, mint NFTs, list assets, and execute purchases directly against deployed smart contracts.
- Implemented decentralized NFT metadata storage using **IPFS/Pinata**, linking token metadata and digital assets to ERC-721 tokens while using OpenZeppelin contracts for standardized NFT functionality and ownership controls.
- Delivered a publicly accessible full-stack application, deploying the React frontend on Vercel and backend services on Render to provide a live, testable Web3 application beyond standalone smart contracts.

3. LayerZero Cross-Chain Protocol (Ethereum ↔ Solana)

Tech Stack: Solidity • Rust • LayerZero V2 • Ethereum • Solana • Anchor • Foundry • React • Ethers.js • Solana Web3.js Links: GitHub • Live Demo

- Engineered a bidirectional LayerZero V2 interoperability protocol connecting Ethereum Sepolia and Solana Devnet, supporting cross-chain application messaging and OFT (Omnichain Fungible Token) transfers between EVM and Solana environments.
- Built a unified MetaMask + Phantom wallet frontend for cross-chain interactions, including a Phantom ALT (Address Lookup Table) compatibility workaround to handle Solana transaction size constraints.
- Implemented LayerZero fee quoting and delivery tracking alongside Solana transaction simulation and replay protection, ensuring reliable and verifiable message delivery across chains.
- Delivered peer validation, message verification, and versioned transaction handling across Solidity and Rust/Anchor components, with end-to-end deployment and testing on Ethereum Sepolia and Solana Devnet.

4. Solana Crowdfunding Protocol

Tech Stack: Rust • Solana • Anchor • PDAs • TypeScript • Wallet Adapter Links: GitHub • Live (Vercel)

- Built a Solana crowdfunding protocol in Rust and Anchor supporting campaign creation, SOL contributions, funding-goal tracking, deadline enforcement, creator withdrawals, and contributor refunds.
- Designed PDA-based campaign and protocol state, using deterministic account derivation to associate campaign data with campaign creators without relying on externally generated private keys.
- Implemented Anchor account constraints and authorization rules to validate campaign ownership, signer permissions, contribution state, and refund conditions at the program level.
- Delivered a TypeScript/Wallet Adapter integration layer for wallet-connected campaign interactions, with tests covering successful funding, failed campaigns, and refund behavior.

INTERNSHIPS

Blockchain Developer

Leafnry Web Pack

May 2025 - Jul 2025

Remote

- Assisted in building Web3 dApps integrating smart contracts with frontend and backend systems.
- Worked with ERC-20 and ERC-721 token standards and participated in smart-contract security testing and reviews.

Blockchain Developer

Estatos

Jan 2025 - Feb 2025

Remote

- Developed and optimized smart contracts for blockchain operations.
- Worked on tokenization and DeFi protocols, contributing to secure and efficient smart-contract deployments.
- Collaborated with the tech team to enhance solutions while demonstrating problem-solving and a growth mindset.

Solidity Developer

Nexus Network

Jul 2024 - Dec 2024

Remote

- Developed bridge contracts for relayer-based cross-chain transactions.
- Strengthened smart-contract security and optimization practices.
- Actively participated in team collaborations and implemented feedback for continuous improvement.

EDUCATION

Post Graduate Diploma in Blockchain

IIIT Bangalore & upGrad

Mar 2023 - Mar 2024

Bengaluru, IN

- Completed advanced coursework covering JavaScript, Node.js, Ethereum smart-contract development, and Hyperledger Fabric.
- Developed Ethereum-based smart-contract architectures for decentralized applications including DeFi systems.
- Designed system architecture for the UPSTAC healthcare application, focusing on scalable and secure blockchain integration.
- Explored blockchain-based KYC systems, Hyperledger Fabric network setup, and cross-chain frameworks.

Dual Degree (B.Tech + M.Tech), Energy Science & Engineering

Indian Institute of Technology Bombay (IIT Bombay)

Jul 2014 - Jul 2019

Mumbai, IN

- Secured 61%.

ADDITIONAL INFORMATION

- Languages:** Fluent in Hindi (Native) and English (Professional)
- Interests:** Reading and Cycling